



# Homework #1

## Exercise 1

Design  $\underline{A} \in \mathbb{R}^{2 \times 2}$ ,  $\underline{P} \in \mathbb{R}^{2 \times 2}$  for

$$\dot{x} = A(x - x^*) = Ax$$

Condition for G.A.S.

$$\underline{A^T P + P A = Q}, \quad Q = Q^T < 0$$

a)  $\underline{Q = qI}$ ,  $\underline{q < 0}$

$$\underline{A^T P + P A = qI}, \quad q < 0$$

Design #1

$$\underline{P = I} \quad \text{or} \quad P = \underline{pI} \quad \text{with } p > 0$$

$$A^T P + P A = \underline{A^T + A = qI}$$

$$= 2A = qI$$

$$\Rightarrow A = \frac{1}{2} q I = \begin{bmatrix} \frac{q}{2} & 0 \\ 0 & \frac{q}{2} \end{bmatrix} = \underline{\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}}$$

Design #2

$$A^T P + P A = 2pA = qI \quad \star$$

Design #3

$$P = \begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix} \quad \rightarrow \quad \underline{A = \frac{q}{2p} I}$$

$A^T$

Design #3

$$P = \begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix}$$

$$A^T P + P A = q I$$

$$\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix} + \begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix} \begin{bmatrix} a_1 & a_3 \\ a_2 & a_4 \end{bmatrix} = \begin{bmatrix} q & 0 \\ 0 & q \end{bmatrix}$$

$$a_2 = a_3 = 0$$

$$a_2 = -a_3 \quad q < 0$$

$$a_1 p_1 + a_1 p_1 = q$$

Ex. 1.b  $Q = Q^T = \begin{bmatrix} q_1 & q_2 \\ q_2 & q_1 \end{bmatrix} < 0$

$$\det(Q - \lambda I) = \det \begin{pmatrix} q_1 - \lambda & q_2 \\ q_2 & q_1 - \lambda \end{pmatrix} = (q_1 - \lambda)^2 - q_2^2 = 0$$

$$q_2^2 = (q_1 - \lambda)^2 \Rightarrow \boxed{\lambda = q_1 \pm q_2}$$

$$\boxed{q_1 = -10, q_2 = 2}$$

$$Q = \begin{bmatrix} -10 & 2 \\ 2 & -10 \end{bmatrix}$$

$$Q = \begin{bmatrix} -10 & 2 \\ 2 & -10 \end{bmatrix}, \text{ Find } A, P \text{ s.t.}$$

$$A^T P + P A = \begin{bmatrix} -10 & 2 \\ 2 & -10 \end{bmatrix}$$

$$P = I$$

$$A^T + A = \begin{bmatrix} -10 & 2 \\ 2 & -10 \end{bmatrix}$$

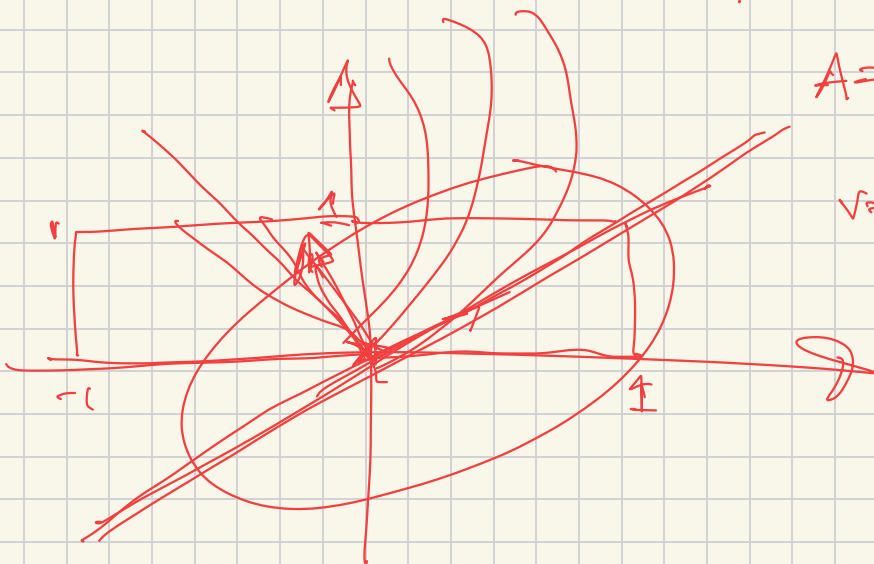
$$\begin{bmatrix} a_1 & a_2 \\ a_2 & a_1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -10 & 2 \\ 2 & -10 \end{bmatrix}$$

$$A = \begin{bmatrix} -5 & 1 \\ 1 & -5 \end{bmatrix}$$

$$P = I$$

$$A = V \Lambda V^T$$

$$V = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix}$$



# Exercise #2

Design  $A_1, A_2 \in \mathbb{R}^{2 \times 2}$  s.t.

$$\dot{x}^* = 0, b = -Ax^* = 0$$

$$\dot{\hat{x}} = \sum_{p=1}^2 \sigma_p(x) A_p x$$

$$V(x) = (x - x^*)^T (x - x^*)$$

$$A_p^T + A_p < 0$$

$$\begin{cases} \sigma_1(x) = 0.7 \\ \sigma_2(x) = 0.3 \end{cases}$$

$$\dot{x} = 0.7 A_1 x + 0.3 A_2 x$$

$$\begin{cases} A_1^T + A_1 < 0 \\ A_2^T + A_2 < 0 \end{cases}$$

$$A_1 = \begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}, A_2 = \begin{bmatrix} -1 & -2 \\ 0.5 & -1 \end{bmatrix}$$

# Exercise #3

$$\gamma_1(x) = 0.5, \gamma_2(x) = 0.5, P = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, Q = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\dot{X} = \sum_{k=1}^2 \gamma_k(x) A_k x, \quad \overline{X^* = 0}$$

$$A_{\text{sum}} = 0.5 A_1 + 0.5 A_2$$

$$A_{\text{sum}}^T P + P A_{\text{sum}} = Q$$

$$A_{\text{sum}}^T \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} A_{\text{sum}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \leftarrow 0$$

$$2 A_{\text{sum}}^T + 2 A_{\text{sum}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$2 A_{\text{sum}} = \begin{pmatrix} -0.5 & 0 \\ 0 & -0.5 \end{pmatrix}$$

$$A_{\text{sum}} = -\frac{1}{4} I$$

$$A_{\text{sum}} = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}$$

$$A_{\text{sum}} = \begin{pmatrix} a_1 & a_3 \\ a_4 & a_2 \end{pmatrix}$$

$$A_{\text{sum}} = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}$$

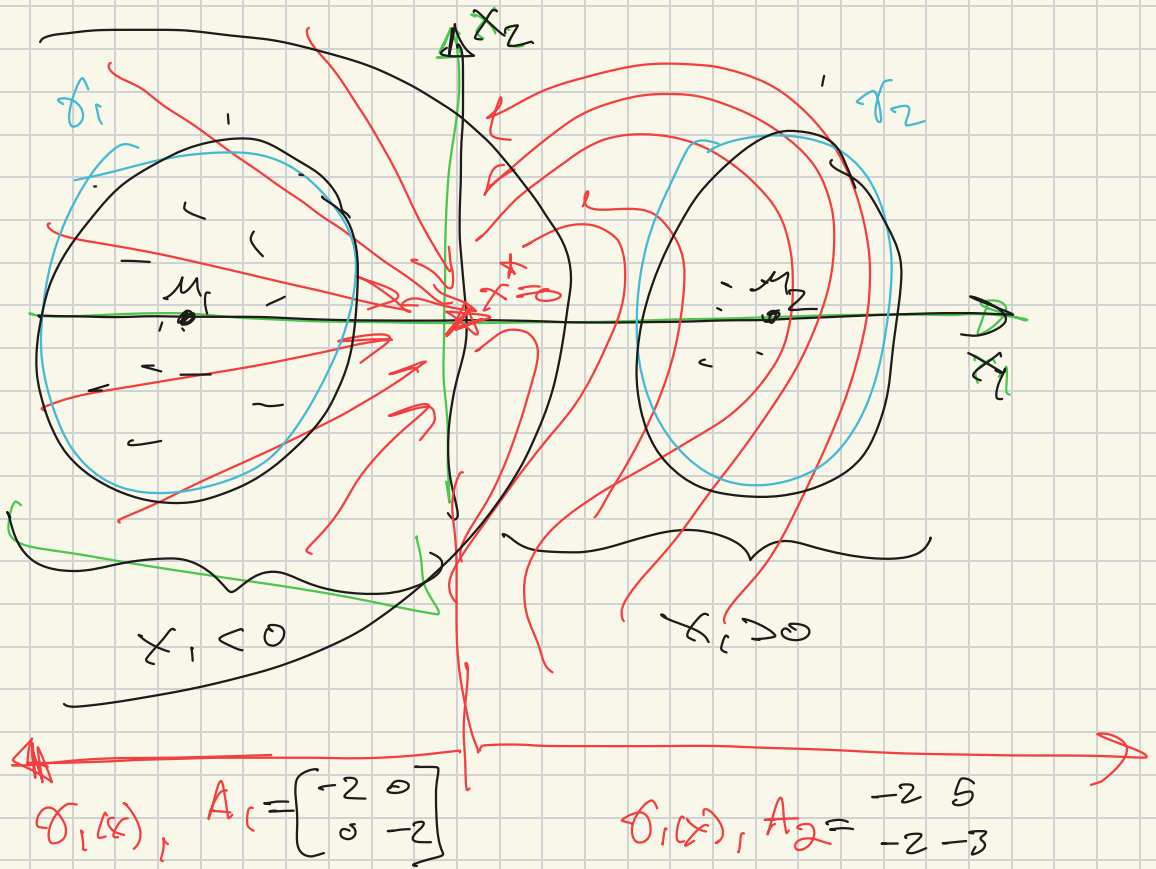
$$0.5 A_1 + 0.5 A_2 = -\frac{1}{4} I$$

$$A_1 + A_2 = -\frac{1}{2} I$$

$$A_1 = \begin{bmatrix} 1 & -2 \\ 4 & 0.5 \end{bmatrix}, A_2 = \begin{bmatrix} -1.5 & 2 \\ -4 & -1 \end{bmatrix}$$

# Exercise #5

Design  $P, A_1, A_2 \in \mathbb{R}^{2 \times 2}$  s.t.  $X^* = \emptyset$ , free-choice of  $Q$

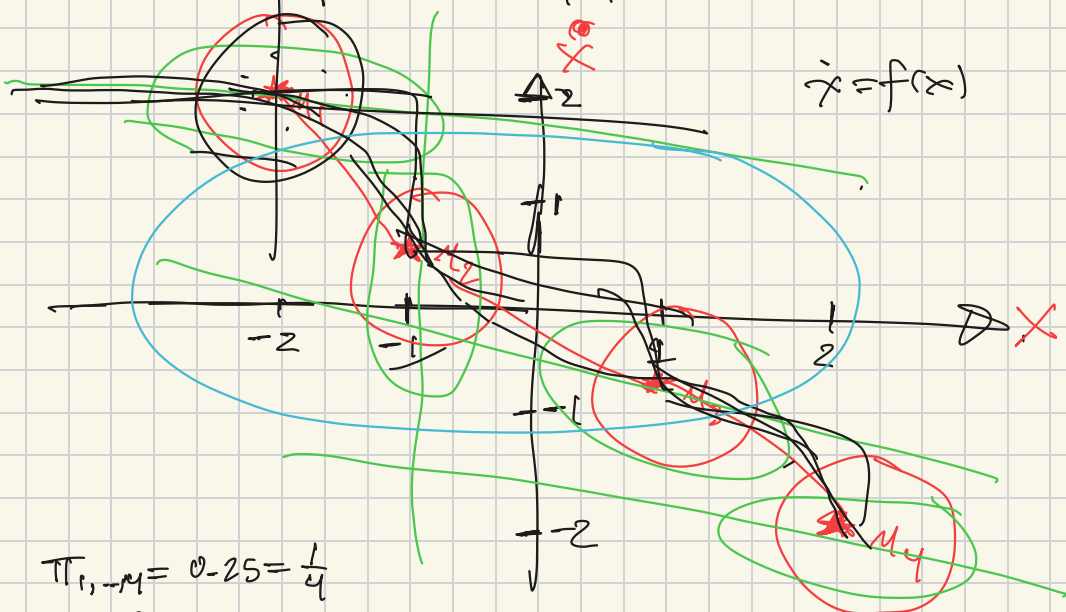


$$\Theta_x \left\{ \begin{array}{l} \pi_1, \pi_2, \overline{\pi_{112} = 0.5} \\ \mu_1, \mu_2, \mu_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \mu_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \varepsilon_1, \varepsilon_2, \varepsilon_{112} = \begin{bmatrix} 0.05 & 0 \\ 0 & 0.10 \end{bmatrix} \end{array} \right.$$

# Exercise #4

$$\hat{x}^* = 0$$

$$\hat{x} = E\{p(\hat{x}|x)\} = \sum_{k=1}^K \gamma_k(x) \mu_k$$



$$\pi_{1,1} = 0.25 = \frac{1}{4}$$

$$\mu_1 = \begin{bmatrix} -1 \\ 0.5 \end{bmatrix}, \mu_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}, \mu_3 = \begin{bmatrix} 1 \\ -0.5 \end{bmatrix}, \mu_4 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

$$\Sigma_{1,4} = \begin{bmatrix} 0.25 & 0 \\ 0 & 0.25 \end{bmatrix}, \Sigma_K = \begin{bmatrix} \Sigma_x & \Sigma_{\hat{x}x} \\ \Sigma_{x\hat{x}} & \Sigma_{\hat{x}} \end{bmatrix}$$

$$\Sigma_2 = \Sigma_3 = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}, \Sigma_{2,4} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$



# Comparison of spherical vs. full covariance matrices

